FitBuddy – AI Fitness Plan Generator

GitHub Project Documentation & Setup Guide

A GitHub-ready documentation PDF based on the supplied project files.

1. Project Overview

FitBuddy is a FastAPI wellness-planning demo using Python, Jinja2, SQLite/SQLAlchemy and Google Gemini. It creates seven-day workout plans, nutrition and recovery tips, and feedback-based plan revisions. The supplied README also describes browser speech features, locale selection and visual exercise/meal references.

2. Technology Stack

Component	Technology
Backend	Python + FastAPI
Frontend	HTML + Jinja2 + CSS/JavaScript
Database	SQLite + SQLAlchemy
AI	Google Gemini via google-genai
Validation	Pydantic + pydantic-settings
Server	Uvicorn
Testing	Pytest + HTTPX

3. Project Structure

```
FitBuddy/
■■■ app/
■■■ templates/
■■■ static/
■■■ tests/
■■■ .github/workflows/
■■■ .env.example
■■■ .gitignore
■■■ README.md
■■■ requirements.txt
```

4. Installation

```
py -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
Copy-Item .env.example .env
```

5. Configuration

The local environment file contains sensitive configuration. This PDF intentionally does not reproduce the API key. Configuration variable names present in the uploaded local environment are:

```
GEMINI_API_KEY
GEMINI_WORKOUT_MODEL
GEMINI_TIP_MODEL
DATABASE_URL
ADMIN_KEY
APP_NAME
DEBUG
```

Important: Keep .env private. If the uploaded API key is real and has been exposed outside your private environment, revoke/rotate it and create a replacement key. The supplied .gitignore excludes .env and database files from Git tracking.

6. Run the Application

```
py -m uvicorn app.main:app --reload

http://127.0.0.1:8000/
http://127.0.0.1:8000/health
http://127.0.0.1:8000/docs
```

7. Tests

```
py -m pytest -q
```

8. GitHub Upload Checklist

Upload the application source, templates, static assets, tests, README.md, requirements.txt, .gitignore, .env.example and GitHub Actions workflow. Do not upload .env, fitbuddy.db, test database files, or API credentials.

9. Safety and Scope

The supplied README describes FitBuddy as a wellness-planning demo rather than a medical service. It does not diagnose conditions, prescribe medication, provide calorie targets, or recommend extreme dieting. It specifies lower-volume movement suggestions and non-restrictive nutrition guidance for users under 18.

10. Repository Purpose

This PDF can be stored in the GitHub repository as project documentation, for example: docs/FitBuddy_Project_Documentation.pdf.

Source basis: uploaded README.md, requirements.txt, .env, .env.example and .gitignore. The actual secret value from .env is intentionally excluded.